



Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant**ICON Central Office - Madhapur - Hyderabad**SEC: **Sr. XL-500, Sr.S60(N&S) & LIIT**

JEE-MAIN

Date: 19-12-2025

Time: **09.00Am to 12.00Pm**

GTM-02

Max. Marks: 300

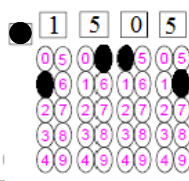
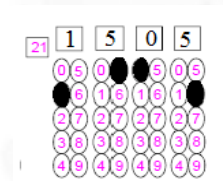
IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black Ball Point Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions**. The maximum marks are **300**.
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics and Chemistry** having **25 Questions** in each subject and subject having **two sections**.
(I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
(II) Section-II contains **05 Numerical Value Type Questions**.

- The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

To cancel any attempted question bubble on the question number box.

For example: To cancel attempted Question 21. Bubble on 21 as shown below

**Question Answered for Marking Question Cancelled for Marking****Marking scheme: +4** for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number:

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Candidate's Signature: _____

Invigilator's Signature: _____

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**Sri Chaitanya**
Educational Institutions**THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1**
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023**JEE MAIN**
2023SINGARAJU
VENKAT KOUNDINYA
Sri Chaitanya
Class 12th Class**300**
RANK**RANK**
1**JEE Advanced**
2023VAVILALA
CHIDVILAS REDDY
Sri Chaitanya
Class 12th Class**341**
RANK**RANK**
1**NEET**
2023BORA VARUN
CHAKRAVARTHI
Sri Chaitanya
Class 12th Class**278**
RANK**RANK**
1**For More Join: @ITTJEE_Advanced**



19-12-2025_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT Jee-Main_GTM-02_Test Syllabus

MATHEMATICS : On ONLY FIRST Year SYLLABUS

PHYSICS : On ONLY FIRST Year SYLLABUS

CHEMISTRY : On ONLY FIRST Year SYLLABUS

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**JEE MAIN
2023**

SINGARAJU
VENKAT KOUNDINYA
Sri Chaitanya
JEE 12th Class

306
100
RANK



RANK

1

**JEE Advanced
2023**

VAVILALA
CHIDVILAS REDDY
Sri Chaitanya
JEE 12th Class

341
100
RANK



RANK

1

**NEET
2023**

BORA VARUN
CHAKRAVARTHI
Sri Chaitanya
JEE 12th Class

700
100
RANK



RANK

1

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**MATHEMATICS****Max Marks: 100****SECTION-I****(SINGLE CORRECT ANSWER TYPE)**

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

1. Let $f(x) = \begin{cases} 1+x; & 0 \leq x \leq 2 \\ 3-x; & 2 < x \leq 3 \end{cases}$, $g(x) = f(f(x))$

LIST-I

LIST-II

A) If domain of $g(x)$ is $[a, b]$ then $b-a$ is

P) 1

B) If range of $g(x)$ is $[c, d]$ then $c + d$ is

Q) 2

C) $f(f(f(2))) + f(f(f(3)))$ is

R) 3

D) $m = \text{maximum value of } g(x) \text{ then } 2m-2$ is

S) 4

Which is the correct match?

1) $A \rightarrow R, B \rightarrow Q, C \rightarrow R, D \rightarrow P$ 2) $A \rightarrow R, B \rightarrow R, C \rightarrow R, D \rightarrow S$ 3) $A \rightarrow R, B \rightarrow S, C \rightarrow R, D \rightarrow S$ 4) $A \rightarrow Q, B \rightarrow P, C \rightarrow S, D \rightarrow R$

2. The number of elements in the exhaustive set of values of x for which

$$f(x) = \sqrt[5]{x^2|x|^3} - \sqrt[3]{x^2|x|} - 1 \text{ is not differentiable is}$$

1) 2

2) 3

3) 4

4) 0

3. Let ABC be a triangle such that $\overline{BC} = \overline{a}$, $\overline{CA} = \overline{b}$, $\overline{AB} = \overline{c}$, $|\overline{a}| = 6\sqrt{2}$, $|\overline{b}| = 2\sqrt{3}$ and $\overline{b} \cdot \overline{c} = 12$, consider the statements

Statement-I : $\left| (\overline{a} \times \overline{b}) + (\overline{c} \times \overline{b}) \right| - |\overline{c}| = 6(2\sqrt{2} - 1)$

Statement-II: $\angle ACB = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

1) Statement-I is false but Statement-II is true

2) Both Statement-I and Statement-II are false

3) Both Statement-I and Statement-II are true

4) Statement-I is true but Statement-II is false

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JEE MAIN**2023**

SINGARAJU

VENKAT KOUNDYIN

Sri Chaitanya

Rank 1st Class

300

100

RANK**1****JEE Advanced****2023**

VAVILALA

CHIDVILAS REDDY

Sri Chaitanya

Rank 2nd Class

341

340

RANK**1****NEET****2023**

BORA VARUN

CHAKRAVARTHI

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Rank 1st Class

200

200

RANK**1**

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4. Let α and β be the roots of $x^2 + \sqrt{3}x - 16 = 0$ and γ and δ be the roots of $x^2 + 3x - 1 = 0$. If $Q_n = \alpha^n + \beta^n$ and $P_n = \gamma^n + \delta^n$ ($n \in \mathbb{N}$), then $\frac{P_{25} - P_{23}}{P_{24}} + \frac{Q_{25} + \sqrt{3}Q_{24}}{2Q_{23}}$ is equal to
- 1) 3 2) 4 3) 5 4) 7
5. Let $a_1, a_2, a_3, \dots$ be in A.P such that $\sum_{k=1}^{12} a_{2k-1} = -\frac{72}{5}a_1, a_1 \neq 0$. If $\sum_{k=1}^n a_k = 0$, then n is:
- 1) 11 2) 10 3) 18 4) 17
6. If $y(x) = \begin{vmatrix} \sin x & \cos x & \sin x + \cos x + 1 \\ 27 & 28 & 27 \\ 1 & 1 & 1 \end{vmatrix}, x \in \mathbb{R}$, then $\frac{d^2y}{dx^2} + y$ is equal to
- 1) -1 2) 28 3) 27 4) 1
7. If the system of linear equations $3x + y + \beta z = 3, 2x + \alpha y - z = -3, x + 2y + z = 4$ has infinitely many solutions, then the value of $22\beta - 9\alpha$ is :
- 1) 49 2) 31 3) 43 4) 37
8. Let the three sides of a triangle are on the lines $4x - 7y + 10 = 0, x + y = 5$, and $7x + 4y = 15$. Then the distance of its orthocentre from the orthocentre of the triangle formed by the lines $x = 0, y = 0$ and $x + y = 1$ is
- 1) 5 2) $\sqrt{5}$ 3) $\sqrt{20}$ 4) 20
9. If $\theta \in \left[-\frac{7\pi}{6}, \frac{4\pi}{3}\right]$, then the number of solutions of $\sqrt{3} \operatorname{cosec}^2 \theta - 2(\sqrt{3} - 1) \operatorname{cosec} \theta - 4 = 0$ is equal to
- 1) 6 2) 8 3) 10 4) 7
10. The value of $\cot^{-1}\left(\frac{\sqrt{1+\tan^2(2)}-1}{\tan(2)}\right) - \cot^{-1}\left(\frac{\sqrt{1+\tan^2\left(\frac{1}{2}\right)}+1}{\tan\left(\frac{1}{2}\right)}\right)$ is equal to
- 1) $\pi - \frac{5}{4}$ 2) $\pi - \frac{3}{2}$ 3) $\pi + \frac{3}{2}$ 4) $\pi + \frac{5}{2}$





11. Each of the angles β and γ that a given line makes with positive y – and z- axes, respectively, is half of the angle that this line makes with the positive x-axis. Then the sum of all possible values of the angle β is
- 1) $\frac{3\pi}{4}$ 2) π 3) $\frac{\pi}{2}$ 4) $\frac{3\pi}{2}$
12. Let the shortest distance between the lines $\frac{x-3}{3} = \frac{y-\alpha}{-1} = \frac{z-3}{1}$ and $\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-\beta}{4}$ be $3\sqrt{30}$. Then the positive value of $5\alpha + \beta$ is
- 1) 42 2) 46 3) 48 4) 40
13. The sum of the cubes of all the roots of the equation $x^4 - 3x^3 - 2x^2 + 3x + 1 = 0$ is
- 1) 32 2) 36 3) 25 4) 64
14. If $f(x)$ is defined on $(0,1)$, then the domain of definition of $f(e^x) + f(\ln|x|)$ is
- 1) $(-e, -1)$ 2) $(-e, -1) \cup (1, e)$ 3) $(-\infty, -1) \cup (1, \infty)$ 4) $(-e, e)$
15. The range of $f(x) = \left[\frac{1}{\sin\{x\}} \right]$ is
(Where $\{ \cdot \}$ is fractional part function and $[\cdot]$ is G.I.F)
- 1) $\{1, -1\}$ 2) $\{0\}$ 3) $\mathbb{N}$ 4) $\mathbb{Z}$
16. Find the minimum value of $f(x) = \frac{4}{x+2} + x, x \in [0, \infty]$
- 1) 2 2) -3 3) -6 4) -9
17. The maximum area of the rectangle whose sides pass through the vertices of a given rectangle of sides a and b is
- 1) $2(ab)$ 2) $\frac{1}{2}(a+b)^2$ 3) $\frac{1}{2}(a^2 + b^2)$ 4) $\frac{1}{2}(a^2 - b^2)$
18. If $A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$ then A^{2025} is _____
- 1) 2A 2) 2I 3) I 4) A





19. The interior angles of a polygon are in A.P. The smallest angle is 120° and common difference is 5° then the number of sides of polygon is
- 1) 6 2) 9 3) 20 4) 7
20. Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = 2\hat{i} + 3\hat{j} - 5\hat{k}$ and $\vec{c} = 3\hat{i} - \hat{j} + \lambda\hat{k}$ be 3 vectors. Let $\vec{r}$ be a unit vector along $\vec{b} + \vec{c}$. If $\vec{r} \cdot \vec{a} = 3$ then $6\lambda =$
- 1) 27 2) 30 3) 25 4) 50

SECTION-II (NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. If $M(r) = \sin^r \theta + \cos^r \theta$, then maximum value of $\left| \frac{M(2025) - M(2023)}{M(2021)} \right| (2024)$ is
- Note: $| \cdot |$ denotes absolute value of real number.
22. Let $f(x)$ is a continuous and thrice differentiable function in $\mathbb{R}$ and given four statements
- S1: If $f(x)$ has a local maximum at $x = a$, then $f''(a) < 0$
- S2: If $f''(a) < 0$ then $f(x)$ has a local maximum at $x = a$
- S3: If $f''(a) = 0$ then $f(x)$ has neither local maximum nor local minima at $x = a$
- S4: If $f(x)$ has neither local maxima nor local minima at $x = a$, then $f'(a) \neq 0$
- Then how many of the given four statement (s) is / are **INCORRECT**?
23. If $S = \{a \in \mathbb{R} : |2a - 1| = 3[a] + 2\{a\}\}$ where $[t]$ denotes the greatest integer less than or equal to t and $\{t\}$ represents the fractional part of t , then $8 \sum_{a \in S} a$ is equal to _____
24. The number of integral values of x for which the function $\sqrt{7x - x^2 - 6}$ is real is _____
25. Let the quadratic curve passing through the point $(-1, 0)$ and touching the line $y = x$ at $(1, 1)$ be $y = f(x)$. Then the x -intercept of the normal to the curve at the point $(a, a+1)$ in the first quadrant





PHYSICS

Max Marks: 100

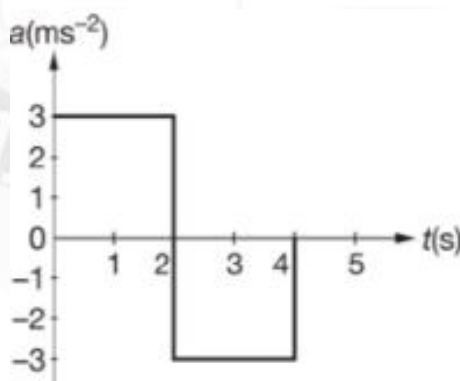
SECTION-I

(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

26. A body rolls down on an inclined plane without slipping. The kinetic energy of rotation is 50% of its translational kinetic energy. The body is:
- 1) solid cylinder
 - 2) Hollow cylinder
 - 3) Ring
 - 4) Solid sphere
27. A body initially at rest is moved along a horizontal straight line by a machine delivering a constant power. The distance moved by the body in time 't' is proportional to:
- 1) $t^{\frac{3}{2}}$
 - 2) $t^{\frac{1}{4}}$
 - 3) $t^{\frac{3}{4}}$
 - 4) $t^{\frac{1}{2}}$
28. Consider a binary star system of star A and star B with masses m_A and m_B revolving in a circular orbit of radii r_A and r_B , respectively. If T_A and T_B are the time period of star A and star B, respectively, then :
- 1) $T_A = T_B$
 - 2) $\frac{T_A}{T_B} = \left(\frac{r_A}{r_B}\right)^{\frac{3}{2}}$
 - 3) $T_A > T_B$ (if $m_A > m_B$)
 - 4) $T_A > T_B$ (if $r_A > r_B$)
29. A particle starts from rest at time $t = 0$ and undergoes acceleration a as shown. The velocity as function of time during the interval 0 to 4 second is indicated in



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RANK

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JEE Advanced
2023VAVILALA
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JEE 12th Class34th
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Rank

RANK

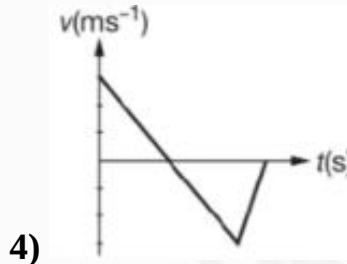
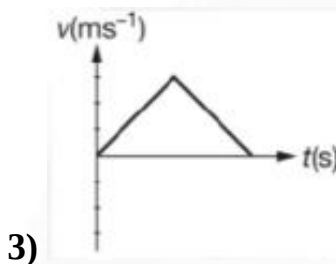
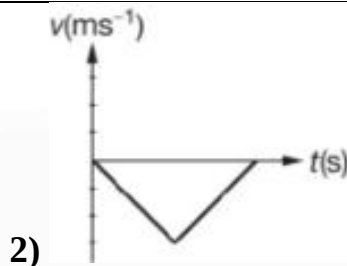
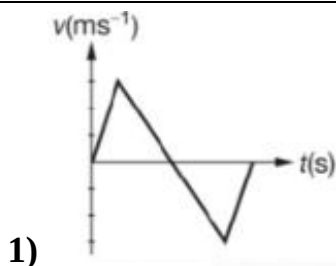
1

NEET
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NEET 12th Class2nd
100%
Rank

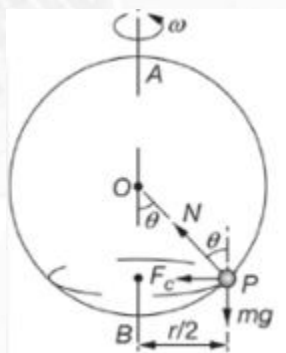
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30. A smooth wire of length $2\pi r$ is bent into a circle and kept in a vertical plane. A bead can slide smoothly on the wire. When the circle is rotating with angular speed ω about the vertical diameter AB, as shown in figure, the bead is at rest with respect to the circular ring at position P as shown. Then the value of ω^2 is equal to

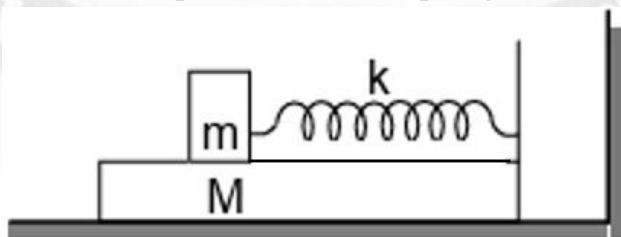


- 1) $\frac{(g\sqrt{3})}{r}$ 2) $\frac{2g}{r}$ 3) $\frac{2g}{(r\sqrt{3})}$ 4) $\frac{\sqrt{3}g}{2r}$
31. A body slides over an inclined plane forming an angle of 45° with the horizontal. The distance x travelled by the body in time t is described by the equation $x = kt^2$, where $k = 1.732$. The coefficient of friction between the body and the plane has a value ($\sqrt{6} = 2.45$ and $g = 9.8 \text{ m/s}^2$)
- 1) $\mu = 0.5$ 2) $\mu = 1$ 3) $\mu = 0.25$ 4) $\mu = 0.75$





32. Statement -I: The total translational kinetic energy of all the molecules of a given mass of an ideal gas is 1.5 times the product of its pressure and its volume.
Statement-II: The molecules of a gas collide with each other and the velocities of the molecules change due to the collision
- if both statements are true and statement 2 is the correct explanation of statement-I
 - if both statements are true but statement 2 is not the correct explanation of statement-I
 - if statement-I is true and statement 2 is false
 - if statement-I is false and statement 2 is true.
33. A thin circular plate of mass M and radius R has its density varying as $\rho(r) = \rho_0 r$ with ρ_0 as constant and r is the distance from its centre. The moment of inertia of the circular plate about an axis perpendicular to the plate and passing through its edge is $I = a(MR^2)$. The value of the coefficient a is
- $\frac{3}{2}$
 - $\frac{1}{2}$
 - $\frac{3}{5}$
 - $\frac{8}{5}$
34. The mass density of a planet of radius R varies with the distance r from its centre as $\rho(r) = \rho_0 \left(1 - \frac{r^2}{R^2}\right)$. Then the gravitational field is maximum at
- $r = \frac{1}{\sqrt{3}} R$
 - $r = \sqrt{\frac{5}{9}} R$
 - $r = \sqrt{\frac{3}{4}} R$
 - $r = R$
35. One main scale division of a vernier calipers is 'a' cm and n^{th} division of the vernier scale coincide with $(n - 1)^{\text{th}}$ division of the main scale. The least count of the calipers in mm is:
- $\frac{10a}{(n-1)}$
 - $\frac{10na}{(n-1)}$
 - $\left(\frac{n-1}{10n}\right)a$
 - $\frac{10a}{n}$
36. A system of two bodies of masses 'm' and 'M' being interconnected by a spring of stiffness k , in its natural length, moves towards a rigid wall on a smooth horizontal surface as shown in figure with a K.E of system 'E'. If the body M sticks to the wall after the collision, the maximum compression of the spring will be:



- $\sqrt{\frac{mE}{Mk}}$
- $\sqrt{\frac{2mE}{(M+m)k}}$
- $\sqrt{\frac{2(m+M)E}{mk}}$
- $\sqrt{\frac{2ME}{(M+m)k}}$





37. A sphere of mass $m_1 = 2\text{kg}$ collides with a sphere of mass $m_2 = 3\text{kg}$ which is at rest. Mass m_1 , after the collision will move at right angle to the line joining centers of two spheres at the time of collision, assuming colliding surfaces are smooth, if the coefficient of restitution is
- 1) $\frac{4}{9}$ 2) $\frac{1}{2}$ 3) $\frac{2}{3}$ 4) $\sqrt{\frac{2}{3}}$
38. Water rises to a height of 10cm in a capillary tube and mercury falls to a depth of 3.42cm in the same capillary tube. If the density of mercury is 13.6 g/cc and the angle of contact of mercury and water are 135° and 0° respectively, the ratio of surface tension of water to mercury is
- 1) 1 : 0.15 2) 1 : 3 3) 1 : 6.5 4) 1.5 : 1
39. A student performs an experiment to determine the Young's modulus of a wire, exactly 2 m long, by Searle's method. In a particular reading, the student measures the extension in the length of the wire to be 0.8 mm with an uncertainty of $\pm 0.05\text{mm}$ at a load of exactly 1.0 kg. The student also measures the diameter of the wire to be 0.4 mm with an uncertainty of $\pm 0.01\text{mm}$.
- Take $g = 10.0\text{ms}^{-2}$ (exact). The Young's modulus obtained from the reading is
- 1) $(2.0 \pm 0.9) \times 10^{11} \text{Nm}^{-2}$ 2) $(2.0 \pm 0.2) \times 10^{11} \text{Nm}^{-2}$
- 3) $(2.0 \pm 0.07) \times 10^{11} \text{Nm}^{-2}$ 4) $(3.0 \pm 0.05) \times 10^{11} \text{Nm}^{-2}$
40. Two moles of ideal helium gas are in rubber balloon at 30°C . The balloon is fully expandable and can be assumed to require no energy in its expansion. The temperature of the gas in the balloon is slowly changed to 35°C . The amount of heat required in raising the temperature is nearly (take $R = 8.31 \text{J/mol.K}$)
- 1) 62J 2) 104J 3) 125J 4) 208J





41. Match List-I with List-II

	List-I		List-II
A	Magnetic Induction	I	$ML^2T^{-2}A^{-1}$
B	Magnetic Flux	II	$M^0L^{-1}A$
C	Magnetic Permeability	III	$MT^{-2}A^{-1}$
D	Magnetization	IV	$MLT^{-2}A^{-2}$

Choose the most appropriate answer from the options give below:

- 1) (A)-(ii), (B)-(iv), (C)-(i),(D)-(ii) 2) (A)-(ii), (B)-(i), (C)-(iv),(D)-(iii)
 3) (A)-(iii), (B)-(ii), (C)-(iv),(D)-(i) 4) (A)-(iii), (B)-(i), (C)-(iv),(D)-(ii)

42. For a body executing S.H.M.:

- (A) Potential energy is always equal to its K.E
 (B) Average potential and kinetic energy over any given time interval are always equal
 (C) Sum of the kinetic and potential energy at any point of time is constant
 (D) Average K.E in one time period is equal to average potential energy in one time period

Choose the most appropriate option from the options given bellow:

- 1) (C) and (D) 2) only (C) 3) (B) and (C) 4) only (B)

43. Heat energy of 184 kJ is given to ice of mass 600 g at $-12^\circ C$, Specific heat of ice is $2222.3 \text{ J kg}^{-1}C^{-1}$ and latent heat of ice is 336 kJ / kg^{-1}

- (A) Final temperature of system will be $0^\circ C$
 (B) Final temperature of the system will be greater than $0^\circ C$
 (C) The final system will have a mixture of ice and water in the ratio of 5 : 1
 (D) The final system will have a mixture of ice and water in the ratio of 1 : 5
 (E) The final system will have water only

Choose the correct answer from the options given below:

- 1) A and D only 2) B and D only 3) A and E only 4) A and C only





44. Given below are two statements:

Statement-I : The mean free path of gas molecules is inversely proportional to square of molecular diameter.

Statement-II : Average kinetic energy of gas molecules is directly proportional to absolute temperature of gas.

In the light of the above statements, choose the correct answer from the option given below:

- 1) Statement -I is false but Statement-II is true
- 2) Statement-I is true but Statement-II is false
- 3) Both Statement-I and Statement-II are false
- 4) Both Statement-I and Statement-II are true

45. A projectile is given an initial velocity of $(\hat{i} + 2\hat{j})\text{ m/s}$, where $\hat{i}$ is along the ground and $\hat{j}$ is along vertical. If $g = 10\text{ m/s}^2$, the equation of its trajectory is:

- 1) $y = x - 5x^2$
- 2) $y = 2x - 5x^2$
- 3) $4y = 2x - 5x^2$
- 4) $4y = 2x - 25x^2$

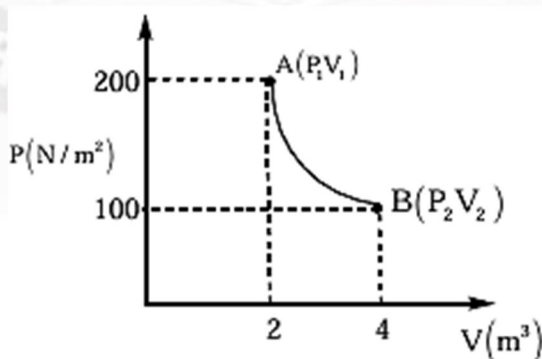
SECTION-II (NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

46. One mole of an ideal gas at 27°C is taken from A to B as shown in the given PV indicator diagram. The work done by the system will be.....J

[Given: $R = 8.3\text{ J/mole K}$, $\ln 2 = 0.6931$] (Round off to the nearest integer)



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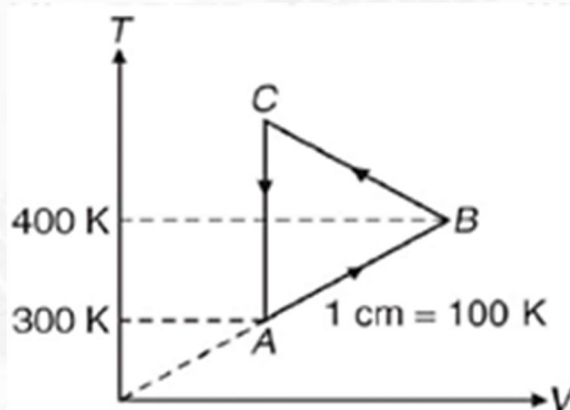
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47. The velocity of upper layer of water in river is 36 kmph. Sharing stress between horizontal layers of water is 10^{-3} Nm^{-2} . Depth of the river is _____ m.
(Coefficient of viscosity of water is 10^{-2} pa-s)
48. Two discs of moments of inertia $I_1 = 4 \text{ kg m}^2$ and $I_2 = 2 \text{ kg m}^2$, about their central axes & normal to their planes, rotating with angular speeds 10 rad/s & 4 rad/s respectively are brought into contact face to face with their axes of rotation coincident. The loss in kinetic energy of the system in the process is _____ J.
49. Figure shows a cyclic process performed on one mole of an ideal gas. A total of 1000 J of heat is withdrawn from the gas in a complete cycle. The magnitude of work done on / by the gas in joule, during the process $B \rightarrow C$ is $10x$ find x . Given $R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$



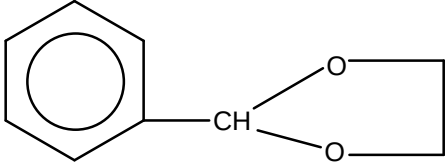
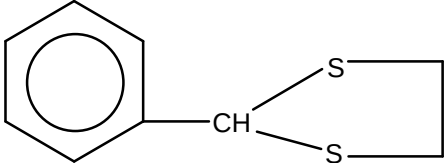
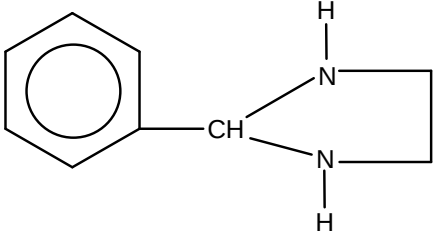
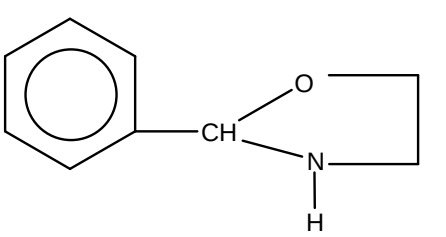
50. One end of a copper rod of length 1 m and area of a cross section $4 \times 10^{-4} \text{ m}^2$ is maintained at 100°C . At the other end of the rod ice is kept at 0°C . Neglecting the loss of heat from the surrounding find the mass of ice, in gram, melted in 1 hr. given $K_{\text{Cu}} = 420 \text{ Wm}^{-1} \text{ K}^{-1}$ and $L_f = 3.36 \times 10^5 \text{ Jkg}^{-1}$



**CHEMISTRY****Max Marks: 100****SECTION-I****(SINGLE CORRECT ANSWER TYPE)**

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

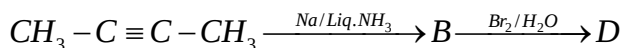
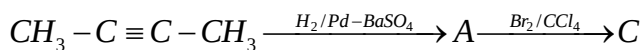
Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

51. 29.9 % (w/w) HCl stock solution has a density of 1.25 g mL^{-1} . The molecular weight of HCl 36.5 g mol^{-1} . The volume (mL) of stock solution required to prepare a 200 mL solution of 0.4 M HCl is $x \times 10^{-2}$. The value of x nearly is _____
 1) 500 2) 600 3) 700 4) 800
52. Nitroglycerine (MW 227.1) detonates according to the following equations,
 $2\text{C}_3\text{H}_5(\text{NO}_3)_3(l) \rightarrow 3\text{N}_2(g) + \frac{1}{2}\text{O}_2(g) + 6\text{CO}_2(g) + 5\text{H}_2\text{O}(g)$ The standard molar enthalpies of information, ΔH_f° for all the compounds are given below
 $\Delta H_f^\circ [\text{C}_3\text{H}_5(\text{NO}_3)_3(l)] = -364 \text{ kJ/mol}$
 $\Delta H_f^\circ [\text{CO}_2(g)] = -393.5 \text{ kJ/mol}$
 $\Delta H_f^\circ [\text{H}_2\text{O}(g)] = -241.8 \text{ kJ/mol}$
 The enthalpy change when 10 g of nitroglycerine is detonated is _____ kJ / mole
 1) -100.5 2) -62.5 3) -80.3 4) -74.9
53. The order of the magnitude of first ionisation potentials of Be, B, N and O is
 1) $\text{N} > \text{O} > \text{Be} > \text{B}$ 2) $\text{N} > \text{Be} > \text{O} > \text{B}$
 3) $\text{Be} > \text{B} > \text{N} > \text{O}$ 4) $\text{B} > \text{Be} > \text{O} > \text{N}$
54. Which of the following compounds has most acidic hydrogen?
 1)  2) 
 3)  4) 
55. In hydrogen atom, the de Broglie wavelength of an electron in the second Bohr orbit is [Given that Bohr radius, $a_0 = 52.9 \text{ pm}$]
 1) 211.6 pm 2) $211.6\pi \text{ pm}$ 3) $52.9\pi \text{ pm}$ 4) 105.8 pm





56. Correct statement about C and D is



1) C is meso & D is Racemic Mixture 2) C is Racemic Mixture & D is meso

3) Both C and D are Racemic mixture 4) both C and D are meso compound

57. The number of neutrons present in the more abundant isotope of boron is 'x'. Amorphous boron upon heating with air forms a product (B_2O_3), in which the oxidation state of boron is 'y'. The value of x + y is ...

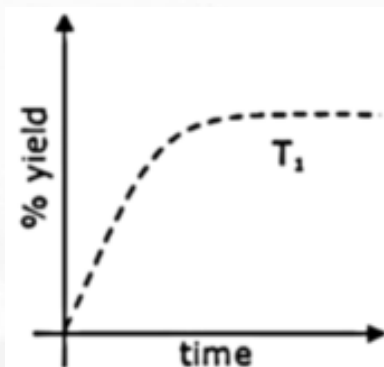
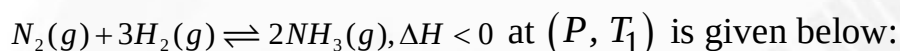
1) 4

2) 6

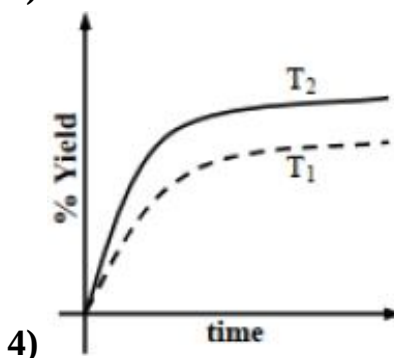
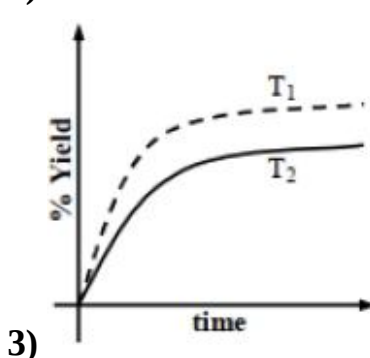
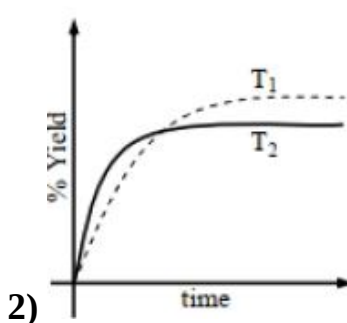
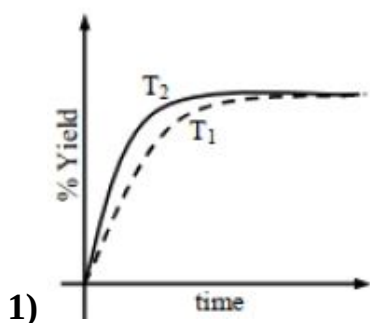
3) 3

4) 9

58. The % yield of ammonia as a function of time in the reaction



If this reaction is conducted at (P, T_2) , with $T_2 > T_1$, the % yield of ammonia as function of time is represented by



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59. Among the following choose the ones with equal number of atoms.

A. 212 g of $Na_2CO_3(s)$ [molar mass = 106 g]

B. 248 g of $Na_2O(s)$ [molar mass = 62 g]

C. 240 g of $NaOH(s)$ [molar mass = 40 g]

D. 12 g of $H_2(g)$ [molar mass = 2 g]

E. 220 g of $CO_2(g)$ [molar mass = 44 g]

Choose the correct answer from the options below

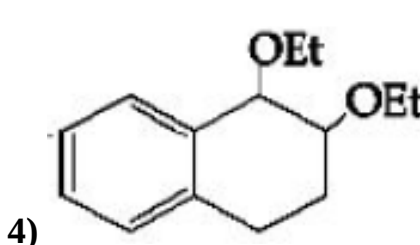
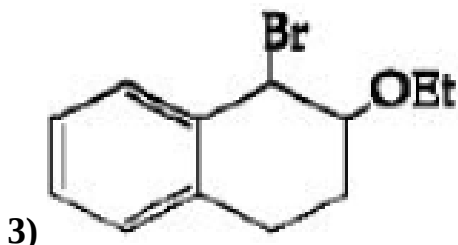
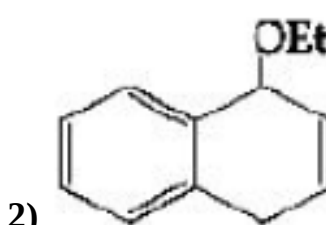
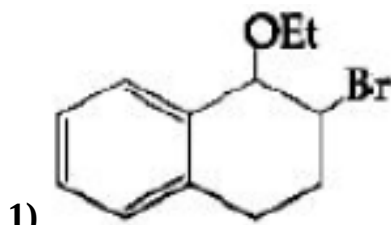
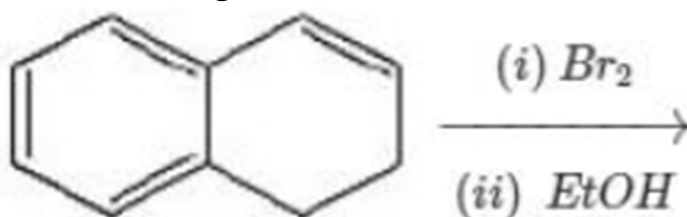
1) A, B and C only

2) A, B and D only

3) B, C and D only

4) B, D and E only

60. The major product of the following reaction is



61. Assertion (A): Shape of chlorine tri fluoride is T – shaped

Reason (R): According Bent's rule more electronegative atom occupies equatorial position in trigonal bipyramidal geometry

1) Both A and R are correct, R is the correct explanation of A

2) Both A and R are correct, but R is not the correct explanation of A

3) A is correct, but R is incorrect

4) A is incorrect, but R is correct





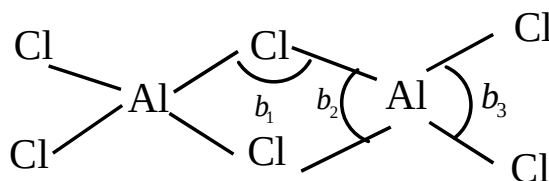
62. Assuming the water vapour to be a perfect gas, the internal energy change when one mole water at 100°C and 1 bar pressure is converted to ice at 0°C is
(given : enthalpy of fusion of ice is 6.00 KJ mol^{-1} heat capacity of water is $4.2\text{ J/g}^{\circ}\text{C}$)
- 1) -7.56 KJ mol^{-1} 2) $-13.56\text{ KJ mol}^{-1}$
3) $+13.56\text{ KJ mol}^{-1}$ 4) $+7.56\text{ KJ mol}^{-1}$
63. The number of ions from the following that are expected to behave as oxidising agent is : Sn^{4+} , Sn^{2+} , Pb^{2+} , Tl^{3+} , Pb^{4+} , Tl^{+}
1) 3 2) 4 3) 1 4) 2
64. Bond order normally gives idea of stability of molecular species. All the molecules viz. H_2 , Li_2 , and B_2 have the same bond order yet they are not equally stable. Their stability order is
1) $\text{H}_2 > \text{B}_2 > \text{Li}_2$ 2) $\text{H}_2 > \text{Li}_2 > \text{B}_2$ 3) $\text{Li}_2 > \text{B}_2 > \text{H}_2$ 4) $\text{B}_2 > \text{H}_2 > \text{Li}_2$
65. Statement -I : Biphenyl is Benzenoid
Statement-II: In Biphenyl, 6π - conjugated electrons are present
1) Both Statement-I & Statement-II are correct
2) Both Statement-I & Statement-II are incorrect
3) Statement-I is correct and Statement -II is incorrect
4) Statement-I is incorrect and Statement -II is correct
66. Assertion: When Inductive and Electrometric effects operate in opposite direction , then electrometric effect predominate
Reason: Electrometric effect is temporary effect
1) Both A and R are correct, R is the correct explanation of A
2) Both A and R are correct, but R is not the correct explanation of A
3) A is correct, but R is incorrect
4) A is incorrect, but R is correct
67. Statement -I
For an adiabatic change in an ideal gas, the relationship between its ΔU and w is $\Delta U = w$
Statement -II
In adiabatic process, $q = 0$





- 1) Both Statement-I & Statement-II are correct
- 2) Both Statement-I & Statement-II are incorrect
- 3) Statement-I is correct and Statement -II is incorrect
- 4) Statement-I is incorrect and Statement -II is correct

68. The bond angles b_1 , b_2 and b_3 in the given structure are



Based on data match the following

List-I (position)

A) b_1

B) b_2

C) b_3

List-II (Bond angle)

I) 101°

II) 79°

III) 118°

- | | | |
|------------|---------|---------|
| 1) A – I | B – II | C – III |
| 2) A – II | B – I | C – III |
| 3) A – III | B – I | C – II |
| 4) A – I | B – III | C – II |

69. Which of the following statements is / are incorrect?

- A) Aniline is separated by steam distillation method from aniline-water mixture
- B) Chloroform is separated from aniline by distillation method
- C) Phosphorus present in the organic compound is not detected by lassaigine's test
- D) In the detection of phosphorus present in the organic compound, the reagent used is Ammonium phospho molybdate

- | | | | |
|-------------|-----------|-------------|-------------|
| 1) A,B only | 2) C only | 3) C,D only | 4) B,D only |
|-------------|-----------|-------------|-------------|

70. Which of the following compounds do not exist

- | | | | | |
|--------------------|------------------|-------------------|--------------------|-------------------|
| A. $[SiCl_6]^{2-}$ | B) $[BF_6]^{3-}$ | C) $[SiF_6]^{2-}$ | D) $[GeCl_6]^{2-}$ | E) $[CCl_6]^{2-}$ |
| 1) A,B,C only | 2) A,B,E only | 3) A,B,D only | 4) A,D,E only | |



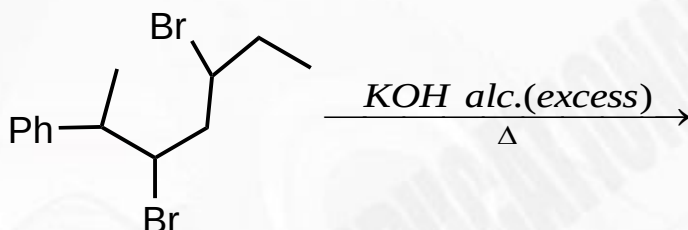


SECTION-II (NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. The number of structural isomers of alkyne possible for the molecular formula C_6H_{10} are
72. Given below are the atomic numbers of some group 14 elements. The atomic number of the element with lowest melting point is :
(6,14,32,50 & 82)
73. The number of sigma bonds present in the major product obtained in the following reaction is



74. The P^H of the resultant mixture obtained by mixing of 10 ml of 0.2 M $Ca(OH)_2$ and 25 ml of 0.1 M HCl is ____.
(Given : $\log(4.28) = 0.631$)
75. The number of '4d' electrons present in Ruthenium (Ru) is 'X' and Osmium (Os) is 'Y'. The value of (X + Y) is ____.
Given : Atomic number of Ru = 44
Atomic number of Os = 76

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